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[22TCSE148]

# FOURTH SEMESTER B.Tech EXAMINATION, APRIL-2024

## STANDARDIZATION

**Time: 3 Hours** Maximum **Marks: 100**

**Instructions:** i. Missing data may be suitably assumed.

**ANSWER ALL QUESTIONS**

**PART-A** **5 X 2=10**

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|---|---|---|---|---|---|
| **1.** | **a.** | Write the areas that government protects by implementing cybersecurity standards. | L1 | CO1 | **02** |
| | **b.** | Expand the following terms as per the computer domain and identify their importance.<br>**i.** ISO<br>**ii.** IEC | L2 | CO1 | **02** |
| | **c.** | Expand the following terms and identify their key differences.<br>**i.** HIPAA<br>**ii.** HITECH | L2 | CO1 | **02** |
| | **d.** | State the role of ITU-T and the OASIS. | L1 | CO1 | **02** |
| | **e.** | Write the importance of PCI DSS standards. | L1 | CO2 | **02** |
| | | **PART-B** | | | **6 X 5=30** |
| **2.** | **a.** | Describe the importance of standardization in cybersecurity. | L2 | C04 | **05** |
| | | **OR** | | | |
| | **b.** | State the drawbacks and challenges occurs to follow standards for cybersecurity. | L4 | C04 | **05** |
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| **3.** | **a.** | Highlight the key differences between ISO 27001 and ISO 27002 standards. | L1 | C04 | **05** |
| | | **OR** | | | |
| | **b.** | How does the relationship between NIST and the Cybersecurity Framework (CSF) contribute to the development and implementation of cybersecurity standards? | L2 | CO3 | **05** |
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| **4.** | **a.** | What are SP 800-53 and SP 800-171, and how do they contribute to cybersecurity standards and compliance? | L4 | C01 | **05** |
| | | **OR** | | | |
| | **b.** | Differences between NIST and other national cybersecurity standards. | L2 | C01 | **05** |
| | | | | | |
| **5.** | **a.** | What are the industry-specific standards in cybersecurity, and why are they important? | L4 | C04 | **05** |
| | | **OR** | | | |
| | **b.** | What is SWIFT's Customer Security Programme (CSP) and how does it work to secure financial transactions? | L3 | C04 | **05** |
| | | | | | |
| **6.** | **a.** | What are the key aspects of implementing new cybersecurity standards, and why are they important for organizations? | L3 | C03 | **05** |
| | | **OR** | | | |
| | **b.** | How do emerging technologies impact cybersecurity standards, and | L2 | C04 | **05** |

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| | | what considerations should organizations keep in mind to adapt to these changes? | | | |
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| 7. | a. | How does international cooperation contribute to the development and implementation of cybersecurity standards, and why is it essential in today's interconnected world? | L4 | C02 | 05 |
| | | **OR** | | | |
| | b. | What are the challenges associated with regulating and standardizing emerging technologies in a rapidly evolving threat landscape? | L2 | C01 | 05 |
| | | **PART-C** | | **6 X 10=60** | |
| 8. | a. | List the benefits of using standards for cybersecurity in government and private sectors. | L1 | C01 | 10 |
| | | **OR** | | | |
| | b. | Compare any two cybersecurity standards and evaluate their strength and weakness. | L2 | C01 | 10 |
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| 9. | a. | Analyze the role and design principles of IEC 62443 in cybersecurity guidelines for industrial automation and control systems (IACS). | L4 | C01 | 10 |
| | | **OR** | | | |
| | b. | What is the relationship between ISO/IEC and other international cybersecurity standards bodies? | L3 | C02 | 10 |
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| 10. | a. | How does the collaboration between ITU-T and OASIS benefit the advancement of international cybersecurity standards? | L4 | C03 | 10 |
| | | **OR** | | | |
| | b. | What role does NIST play in the development and maintenance of the Cybersecurity Framework (CSF)? | L2 | C01 | 10 |
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| 11. | a. | What are the main components of the NIST Cybersecurity Framework (CSF), and how do they contribute to enhancing cybersecurity resilience? | L2 | C02 | 10 |
| | | **OR** | | | |
| | b. | How does the NIST Cybersecurity Framework (CSF) help organizations improve their cybersecurity posture? | L4 | C02 | 10 |
| | | | | | |
| 12. | a. | How do the NIST SP 800 series documents contribute to improving cybersecurity resilience in organizations? | L4 | C02 | 10 |
| | | **OR** | | | |
| | b. | What is the role of the National Institute of Standards and Technology (NIST) in developing cybersecurity standards for the federal government and other organizations? | L1 | C01 | 10 |
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| 13. | a. | How can organizations effectively implement industry-specific cybersecurity standards, such as HIPAA and HITECH standards, to ensure healthcare information security and privacy? | L5 | C02 | 10 |
| | | **OR** | | | |
| | b. | What are the PCI DSS standards, and why are they important for ensuring payment card data security in the financial industry? | L2 | C04 | 10 |